Austin Ulrigg

Curriculum Vitae

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Education

University of Washington September 2023 – June 2026
B.Sc. in Mathematics, Departmental Honors *Seattle, WA*
- GPA: 3.83/4.00

Publications / Preprints

Resolving Pisanski's Conjecture on Quadrilateral Embeddings October 2026
Joint with Alexander Metzger. Preprint, arXiv:2610.13085 *Seattle, WA*
- Proved Pisanski's conjecture on quadrilateral embeddings of regular bipartite graphs by constructing a connected simple d-regular bipartite graph with m vertices in each class and an orientable quadrangulation whenever $m \geq d \geq 4$ and $4 \mid md$.
- Submitted to Journal of Graph Theory.

The Genus of Bipartite Kneser Graphs October 2026
Joint with Alexander Metzger. Preprint, arXiv:2610.03159 *Seattle, WA*
- Determined the orientable genus of the bipartite Kneser graphs $H(h, 2)$ for every prime $h > 3$ with $h \equiv 3 \pmod 8$ by constructing vertex-transitive quadrangular embeddings, which attain the Euler lower bound.
- The construction confirms an infinite family of cases of Pisanski's conjecture on quadrilateral embeddings of regular bipartite graphs.
- Submitted to Journal of Graph Theory.

An Efficient Genus Algorithm Based on Graph Rotations Spring 2024–Winter 2025
Joint work with Alexander Metzger *Seattle, WA*
- Developed a novel algorithm to compute the orientable genus of graphs in $\mathcal{O}(n(4^m/n)^{n/t})$.
- Successfully determined the exact genus of several graphs for which it was previously unknown.
- Published in *Discrete Mathematics* 349 (2026), 115308. doi:10.1016/j.disc.2026.115308. arXiv:2411.07347.
- Integrated into SageMath (`Graph.genus(algorithm='page')`) and used by the House of Graphs database to compute graph genus.

Research Experience

Graph Genera and Minors Fall 2025
Undergraduate Honors Thesis, Supervisor: François Clément *University of Washington, Seattle*
- Authored a comprehensive survey on graph embeddings on surfaces, rotation systems, and genus algorithms (including PAGE).
- Investigated the classification of forbidden toroidal minors and related open problems.
- Full text available: austinulrigg.com/post-thesis.html

Undergraduate Research in Topological Graph Theory Summer 2025–Winter 2026
Advisor: François Clément *Seattle, WA*
- Worked on a project on classifying forbidden minors for the torus, specifically for graphs excluding a K_5 minor.

Washington Experimental Math Lab Researcher March 2024–June 2024; September 2024–December 2024
Advisors: Professor Hadrian Quan; Professor Stefan Steinerberger *Seattle, WA*
- Fall 2024, with Steinerberger: investigated the asymptotic behavior of the dynamical system described in arXiv:2409.08961.
- Spring 2024, with Quan: collaborated with a team to investigate wave propagation on graphs, focusing on spectral analysis of the graph Laplacian, graph products, and path homotopy.

Presentations

A Practical Genus Algorithm May 2025; Nov 2025
UW Undergraduate Research Symposium, Seattle;
Northwest Undergraduate Mathematics Symposium (NUMS), Bothell
- Contributed talks (joint work with Alexander Metzger) on a new approach to genus computation using an $\mathcal{O}(n(4^m/n)^{n/t})$ algorithm, emphasizing theoretical advancements and impacts on computational fields.

Graph Genus Algorithm Feb 2025
Husky Math Club, University of Washington *Seattle, WA*

Euler Characteristic of a Torus and the Utilities Problem July 2024
Centralia High School *Centralia, WA*

- Talk for ∼30 advanced high-school students on surfaces of various genera, Euler's formula, and gave a solution of the utilities problem on a torus.

Honors and Awards

Husky 100 (Nominated by Dept. of Mathematics) *2025*
Math Alliance Predoctoral Scholar Nominee *2024*
Phi Theta Kappa Honor Society (International Honor Society) *2021–Present*
Outstanding Student of the Year Nominee (Centralia College) *2022*
High School Valedictorian (Rank 1/200+, Centralia High School) *2022*
Merit Scholarships Awarded (Total: $12,000) *2022*

Professional Experience

Subject Matter Expert (Contract) Aug 2026 – Present
Mercor *Remote*

- Subject matter expertise for top AI labs through Mercor's expert network; work under NDA.

Mathematics Domain Expert (AI Training) Aug 2023 – July 2024; Aug 2025 – Present
Outlier AI / DataAnnotation *Remote*

- Author and edit complex mathematical prompts and solutions to refine LLM reasoning capabilities. Develop and apply rubrics to assess model performance and categorize content by difficulty.

Outreach and Service

Reading Group Organizer | *University of Washington* 2023 – 2025

- Co-organized and led reading groups in topology, algebra, linear algebra, and complex analysis for 70+ total students, facilitating discussions, grading, and assigning problems.

Teaching Assistant (MATH 209) | *University of Washington* Sep 2024 – Mar 2025

- Assist with ODEs/PDEs (heat, wave, etc.); grade assignments, provide one-on-one support.

Mathematics Tutor (CLUE) | *University of Washington* Sep 2024 – June 2025

- Tutor courses from introductory calculus to abstract algebra.

Teaching Assistant | *Husky Math Academy* Oct 2025 – June 2026

- Provide instruction, building problem-solving skills beyond the standard curriculum.

Math Student Council | *University of Washington* 2024 – 2026

- Council Member: Organize study halls and events; act as liaison between students and faculty.

VP & Recruitment Chair | *Theta Delta Chi Fraternity* 2024 – 2026

- Elected officer for an 80+ member chapter; oversee recruitment and chapter operations.

Online Mathematics Community | *Volunteer Tutor* Jan 2023 – Present

- Authored 100,000+ posts in a community of 200k+ members, facilitating math discussions.

Mathematics Tutor | *Mathnasium* July 2025 – Sep 2025

- Tutored K–12 students, focusing on foundational skills and confidence building.

Solution Notes: Pearls in Graph Theory | *Independent Project* May 2024 – June 2024

- Authored a 12-page solution set for Hartsfield–Ringel's *Pearls in Graph Theory*. [View Notes]

STEM Tutor | *Centralia College* May 2020 – June 2022

- Tutored Calculus I–III, Statistics, and Human Biology; supported center operations.

Graduate Coursework, Languages, and Interests

Coursework: Foundations of Combinatorics (Math 563), Modern Algebra I–III (Math 504–506), Linear Algebra (Math 582–583).

Languages: English (native), Spanish (B1), German (A2).

Hobbies: Volleyball (especially beach), Hiking, Dogs, Powerlifting, Reading